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Experimental Design and Data Analysis for Array Comparative Genomic Hybridization

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ABSTRACT

Array comparative genomic hybridization (aCGH) is a technique for measuring chromosomal aberrations in genomic DNA. With the availability of high-resolution microarrays, detailed characterization of the cancer genome has become possible. In this review, we discuss several issues in the generation and interpretation of aCGH data, including array platforms, experimental design, and data analysis. Due to the complexity of the data, application of appropriate statistical methods is crucial for avoiding false positive findings. We also describe integration of copy number data with other types of data to identify functional significance of observed aberrations.

INTRODUCTION

Chromosomal aberrations play an important role in the pathogenesis and progression of cancer. Multiple copies of an oncogene may be present in the genome due to amplification or a tumor suppressor may no longer be present due to deletion. Aberrations can also target another factor downstream in a pathway. Instead of a gene, aberrations may affect other regulatory elements. A deletion may, for instance, disrupt proper function of a microRNA (1). Copy number aberrations have been studied most extensively in cancer, but they are important in a number of other phenotypes. Importantly, a recent study found that there is a surprisingly large amount of variability in the DNA copy number in the normal population (2). This suggests that the genome-wide landscape of DNA copy number variation is complex, and many aspects of its functional significance are yet to be discovered (3).

A high-resolution characterization of the copy number in the cancer genome has become possible with array comparative genomic hybridization (aCGH). This is a technique for copy number estimation in which cancer DNA and normal DNA are differentially labeled and hybridized on a microarray that contains hundreds of thousands of probes (4, 5). A scanner quantifies the amount of signal from the two labels at each position on the array, and the ratio between the cancer and the normal samples is calculated for each probe. These values from all the probes are then mapped to their chromosomal locations, and the copy number profile is estimated along the genome.

Recent advances in microarray technology have increased the resolution of the platform dramatically while lowering the cost of experiments. This has led to the widespread use of the technology in virtually every cancer type and is likely to have an important clinical application (6). In this review, several key aspects of a successful aCGH study will be described, including choice of microarray platform, experimental design, and statistical analysis. Key steps and ideas in data analysis will be described, as one of the biggest obstacles in taking full advantage of this technology has become the analysis and interpretation of the data. Experimental validations of potentially interesting regions are costly and time-consuming, and thus careful analysis is necessary to find functionally relevant regions reliably. Importance of combining copy number data with other data types, such as gene expression, microRNA expression and methylation profiles, will also be discussed, as such integrative analysis is most likely to yield important biological insights.

Keywords: DNA copy number, CGH, chromosomal aberration, microarrays.
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EXPERIMENTAL DESIGN

The popularity of aCGH is partially due to the increased spatial resolution afforded by a large number of probes on the microarray platform. The first array-based platform utilized BAC (bacterial artificial chromosome) probes, which are generally several hundred base pairs in length and have a median resolution of several mega-bases (4). These long probes have higher specificity than shorter ones, and each probe gives relatively accurate measurements. The resolution of these arrays is sufficient for many applications in which the aberration of interest is large.

Oligonucleotide probes are shorter but provide finer resolution. The more recent oligonucleotide platforms now contain hundreds of thousands probes, in contrast to the BAC arrays that typically have 2000–3000 probes. For instance, the arrays from Agilent contain 244,000 and those from Nimblegen contain 385,000 probes. The next generation arrays of both platforms, available in 2008, contain more than a million probes. The overall quality of the data depends not only on the probe density but also on the quality of the probe design (GC-content, sequence uniqueness), manufacturing technique (probe length, hybridization chemistry), and the level of expertise in carrying out the experiment. But, higher density arrays clearly allow for more detailed mapping of the chromosomal alterations.

All aCGH arrays (not including the SNP arrays discussed below) are two-channel, meaning that a control (reference) sample is hybridized on the array along with the sample of interest. Hence, the log-ratio of 0 in the output corresponds to no change in the copy number compared to the control at that location. The log of base 2 is usually used so that the log-ratio of 1 and -1 corresponds to twice or half as many copies, respectively, in the sample compared to the control.

What to choose for the control channel is an important issue and should be considered well before sample collection begins. In the best case, normal DNA from the same individual is used as a control. The advantage of this approach is that copy number polymorphisms (patient-specific variation not related to the disease) are present in both samples and therefore are not found as aberrations. The difficulty, however, is that normal DNA is difficult to obtain for certain tissues (e.g., brain). A normal tissue is not always obtained in surgery, and, even when it is collected, 'normal' tissue near the tumor may not be normal. In some cases, DNA from the blood, if available, is used as a substitute. If a generic reference sample is used as a control for all samples, both copy number polymorphisms and cancer-related aberrations are found together, and it is difficult to distinguish between them. One solution to this is to look up a database of genomic variants to distinguish copy number polymorphisms from true aberrations. But the list of variants in such databases is far from complete, due to the limited number of samples used and the low resolution platform used. Thus, the likelihood of detecting a false positive is increased when a proper control is not available.

In addition to the availability of appropriate controls, those samples should ideally have well-annotated clinical data. This will allow the investigator to do correlative analysis between any DNA alterations and clinical phenotypes. The most common is survival analysis, in which DNA alterations that may be related to patient survival times are identified. Many other variables, such as age, gender, treatment, drug response, and tumor size can be tested for correlation to the features found in the copy number profiles (7). In practice, many have found that obtaining clinical data in a retrospective study is very time-consuming and is likely to lead to incomplete data. Thus, identifying samples with good clinical annotations or planning a prospective study with appropriate provisions is critical for deriving maximal amount of information from aCGH data (8).

Another important consideration in experimental design is whether other studies may be conducted in the future with the samples from the same set of patients. In most previous profiling studies, investigators were initially interested in a single data type and obtained just enough materials for their study. Thus, when new technologies are developed or new platforms become available, they could not generate new data on the same set of patients. For instance, while hundreds of studies have been done on gene expression and copy number of tumors, only a handful has both types of data on the same patients. This lack of "paired" data sets has seriously hampered integrative analyses. While it is possible to correlate average of expression levels with averages of copy numbers in an unpaired study, it is much less effective than comparing at the level of individuals from a paired case. Therefore, if a future study using RNA or DNA is conceivable, it would be wise to collect the material in advance. Formalin-Fixed Paraffin-Embedded (FFPE) samples have not been used often so far in expression or copy number studies because nucleic acids isolated from such sources are generally degraded. However, techniques to amplify and label such samples are being developed.

Besides the above aCGH platforms, single nucleotide polymorphism (SNP) arrays can also be used to estimate copy number. These are mostly single-color arrays, in which either a tumor or a normal sample can be hybridized. The most common platform for this has been the Affymetrix (9) and Illumina (10) SNP arrays. These arrays initially contained 10 K probes, but subsequent versions with 100 K and 500 K appeared soon afterwards. The latest 6.0 arrays contain 1.8 million markers, one half for SNPs and the other half for copy number variations. The latest Illumina BeadChips also contain more than a million SNPs and a number of markers for copy number variant regions. The SNP arrays have the important advantage of measuring copy number changes and loss of heterozygosity (LOH) simultaneously (8). Recently, a number of algorithms have been proposed to estimate allelic-specific copy numbers (11, 12), which would be important for cancer cells and for copy number-variant regions of normal cells. The disadvantage is that the probe design and placement is not optimal for copy number estimation, and the resulting data tend to be less accurate at the level of individual probes. How to best estimate copy numbers from these arrays is still under development and is not discussed here. Which one is preferable for copy number estimation between the copy number arrays and SNP arrays is still under debate.

The question of minimum sample size required for a study has not been addressed rigorously in the literature so far. The answer depends on multiple factors such as the array resolution, the number and sizes of aberrations expected, heterogeneity of the samples, and the aim of the study. Typical studies have a minimum of 20-30 samples and a few have more than 100-200. The key principle that holds true in any array-based study is the

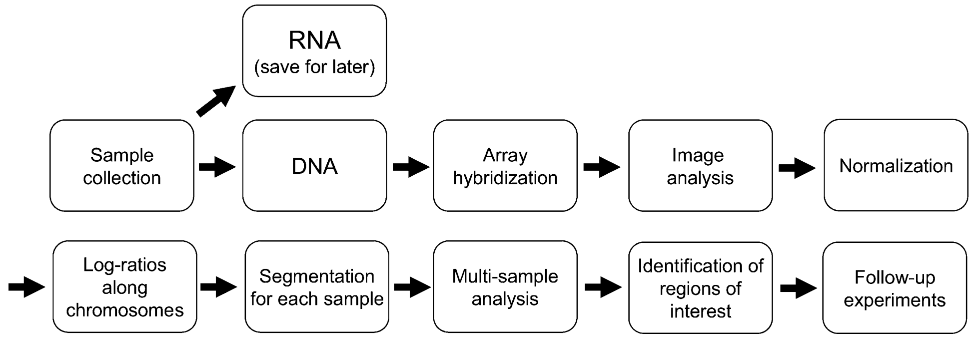

Figure 1. Analysis pipeline for array comparative genomic hybridization. The steps shown in the first row have been examined extensively in the context of gene expression studies. The steps shown in the second row are specific to aCGH studies. If RNA is available from the same patients, it may be prudent to save them for expression profiling, as joint analysis of DNA and RNA is more powerful.

trade-off between the size of the study and the effort required in validation. A large study will have a lower rate of false positive results and thus attempts at validation will be more successful; a smaller study will have a higher rate of false positives and the validation part will be less successful.

DATA ANALYSIS

The key steps in data analysis are outlined in Figure 1. Basic processing of aCGH data is identical to that of the two-color gene expression arrays, for which analytical techniques have been studied extensively and numerous tools are already available (13). To summarize, an image processing algorithm quantifies the strength of the signal from each of the channels and a log-ratio is obtained for each probe. Ideally, the distribution of the ratios should not depend on the signal intensity, but this is often not the case. This dependency comes from the differences in the dyes that are used to label the samples. To reduce this problem, a normalization procedure is generally applied (14). This is often called "lowess" normalization, referring to the name of a statistical technique for the curve-fitting step.

Once log-ratios are obtained for each probe, the main challenge is to scan along the chromosomes to identify regions of copy number aberrations. Intuitively, if there are multiple probes with high log-ratios in a row, that is likely to be a region of amplification; likewise, a string of negative log-ratios indicates a potential region for deletion. The larger the magnitude of those log-ratios and the number of such probes, the stronger the evidence is for amplification or deletion. In practice, measurements from many of the probes are noisy and give errant values. There may be, for instance, negative values scattered among positive values in a region of amplification.

Given these complications, the process of identifying aberrations is best carried out with a statistical method. A number of such 'segmentation' methods are available in the literature, and many are being developed or refined. The most popular approach currently is Circular Binary Segmentation (15). This is a modification of a well-known method for the "change-point problem" in statistics and thus has the advantage of having a solid theoretical foundation. The basic idea is to consider each potential segment of aberration and compare the distribution of the log-ratios in the segment to those outside using, for instance, the t-test. This procedure can be performed recursively until no such segment is found, according to the user-defined threshold p-value. One disadvantage of this method has been the slow speed, but it has been alleviated to some extent with a modified version (16). The method still does not consider distances between adjacent probes in its statistical evaluation, but how much information is lost due to this is not clear.

There are a number of other algorithms that are frequently used, but we defer to other places for the details on those methods (17, 18). We note that no statistical method is likely to be the best method for every platform or every type of profile, and it would be worthwhile for the user to visually inspect how well an algorithm performs for a given data set. A statistical algorithm can be used to scan through the genome quickly to identify regions of potential interest, but in many cases it does not outperform the human eye.

Once individual profiles are 'segmented,' the next step is to combine the results from multiple samples. Perhaps the simplest, yet fairly effective, approach is simply to find the 'average' profile as defined by the average value of all measurements for each probe (19). An implicit assumption in this case is that an aberration is given weight proportional to its magnitude, e.g., an amplification that occurs in one sample with log-ratio of 3 is as important as one that occurs in three samples with log-ratio of 1. An alternative approach is to focus more on the frequency of the aberrations and less on the magnitude of each one (20). The strength of this approach is that infrequent aberrations with relatively low magnitude that may be missed in the first approach would be captured. In general, both approaches are simple, and each has its own merit. It is up to the investigator to choose how to balance magnitude and amplitude.

More sophisticated methods also exist for deriving a consensus profile (21, 22). The idea behind these methods is to account for the block structure present in each individual profile, rather than considering each probe as independent as done

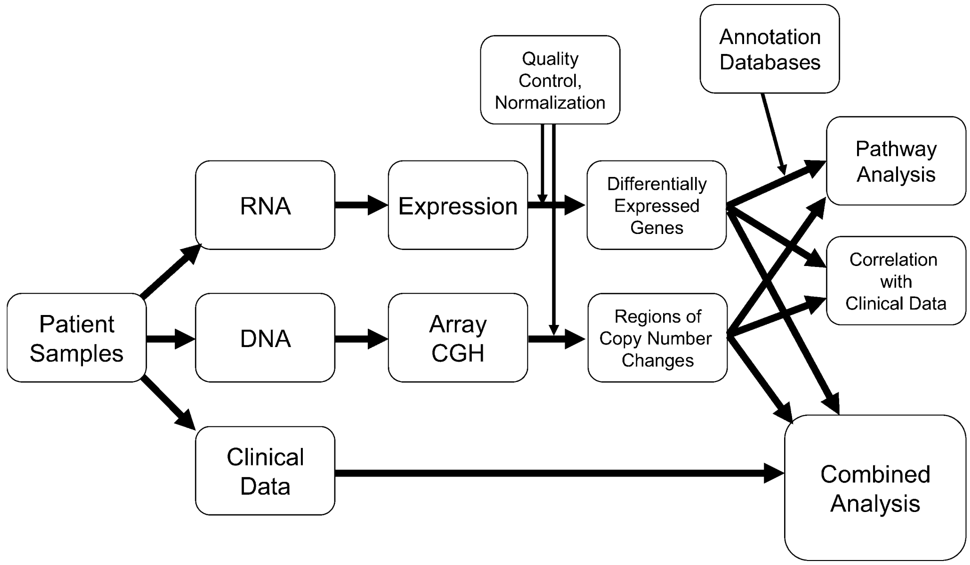

Figure 2. Overview of integrative analysis involving DNA, RNA, and clinical data. Once mRNA transcript and copy number data have been processed, they should be combined to identify features that are likely to be functionally relevant. This includes pathway analysis and correlation analysis involving any pair or all of mRNA, copy number, and clinical data. Other data types, such as microRNA and mutation data, should also be incorporated as they become available.

in the two simple methods above. These methods should give some improvement in terms of identifying the common region of aberrations, but how much is gained through these approaches over the much simpler ones is not clear.

INTEGRATIVE ANALYSIS

While a global view of the copy number changes is helpful for finding a set of potentially important genes, such a set may be too large to provide an obvious next step. In a typical cancer genome, large stretches of genes and even whole chromosome arms may be amplified or deleted. In such cases, it is not clear which genes should be chosen for further study. One way to narrow the list of genes is to look for tumor suppressor genes or oncogenes in the regions of loss or gain, respectively. Databases containing such genes are available publicly. A more unbiased approach is to integrate copy number with other types of data, as outlined in Figure 2. When clinical data are available, for instance, one can find genes with expression levels that are highly correlated with patient survival, treatment history, histological diagnosis, or other such clinical variables. The statistical method used depends on the variable: for a continuous clinical variable, the Pearson or Spearman correlation coefficient; for a discrete variable, a chi-square test or analysis of variance (ANOVA); and for a censored variable, linear regression with the Cox model may be appropriate. The set of genes obtained using this analysis can be intersected with those showing copy number aberrations to obtain genes that are more likely to be functionally relevant.

Perhaps the most obvious high-throughput data type to be integrated with copy number is gene expression (23, 24). But even in this case, the relationship can be complex, as many have observed that the correlation between the two data types is lower than what one would expect at first. For instance, analysis of copy number and expression profiles from the same breast cancer cell lines showed that only 44% of highly amplified genes (>2.5 in copy number ratio) were up-regulated (25). Most other studies have obtained even lower agreements (26).

In combining expression and copy number, the main idea is that those copy number aberrations that result in differential expression of the genes contained within the same aberrations are more likely to be relevant. That is, even if there is a copy number change, it is of less interest if it does not directly result in the change of expression for the same genes (exception noted below). Conversely, there will be many differentially expressed genes, but they are less likely to be upstream events if the differential expression was due to another factor upstream rather than due to a copy number change.

How to perform this analysis depends on whether one has *paired* data in which the two data types are available for each of the patients or *unpaired* data where the data types were obtained from different sets of patients. For paired data, analysis is more direct and much more effective. One can simply scan the genome and look for segments in which expression and copy number are highly correlated across the samples. For high-resolution platforms, this step should be applied after the profiles are segmented as described above. To increase the likelihood of finding true positive regions, one can look for cluster of probes that display high correlation, rather than isolated cases.

When the data are unpaired, one can proceed in one of two ways. One approach is to identify genes that are differentially expressed from expression data and then examine the copy number profiles at those locations. To make this more effective, one can find clusters of genes that are differentially expressed, rather than a single such gene, with the assumption that the region of aberration covers multiple genes. While this assumption is not always valid and micro-aberrations may be missed, it provides a reasonable first step for narrowing down the candidates. Another approach is to first identify regions of copy number aberrations that occur in multiple samples. One can then search for differentially expressed genes in those regions. In both approaches, a further filtering can be done using other data types available, for instance, by using clinical data to select features that are related to patient survival times. This type of analysis was performed previously (27). We, however, note that paired analysis is much more powerful and that one should therefore plan ahead at the beginning of the project to collect appropriate material.

Even if copy number aberrations do not result in changes in expression of those genes, it does not necessarily mean that the aberrations are not important. The aberrations can contain other regulatory elements. A lost transcription factor binding site, for instance, can affect transcription of a gene away from the aberration and thus may not be easily detected in the above analysis. An over-expressed or deleted microRNA may affect the expression of target genes downstream in a pathway. Therefore, it is important to compare copy number profiles with any other data types available. We have described a procedure for correlating copy number with gene expression above, but this type of analysis can be extended to other data types.

One of the main difficulties in these analyses is the difference in the resolution of the platforms. For expression and copy number comparison, for example, the number of probes and their locations are different. The expression platform will have probes on the genes whereas the copy number platform will have probes more evenly distributed along the genome. Even when the same region is covered, a segmented genomic region defined by one probe in one platform may include several probes in the other platform. If the measurements by those multiple probes are highly variable, some type of averaging may need to be performed.

While the number of genes expression and aCGH studies has increased rapidly, it has been difficult to integrate these because of the different patient samples on which these studies were conducted. Recognizing this, the National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI) have launched an ambitious large-scale project in which the latest platforms for every data type are used to profile the same set of patients. The Cancer Genome Atlas (TCGA) project was started in 2006, with the aim of profiling hundreds of samples from three tumors types: glioblastoma, lung, and ovarian. The platforms included in the project are Agilent 244 K CGH arrays, Affymetrix SNP 6.0 arrays, Illumina 550 SNP arrays, Affymetrix U133A expression arrays, Affymetrix Exon arrays, Agilent expression arrays, Agilent microRNA arrays, and Illumina GoldenGate DNA methylation arrays. Multiple institutions with a track record in particular platforms were selected to carry out the profiling experiments. Criteria for the samples chosen for this project include the availability of a large amount of DNA and RNA to be divided up among the different centers and the availability of a full set of clinical annotations. Regions of interest obtained from analysis of this data set are being sequenced for mutations at three of the national sequencing centers. As this data set was intended to be a public resource, all data from the project are made publicly available (http://cancergenome.nih.gov) as soon as samples are processed and have gone through a quality control step. Investigators are encouraged to use this data set and are free to publish results based on them after an initial publication from the TCGA consortium.

CONCLUSION

Array CGH profiling provides a genome-wide view of DNA aberrations in the cancer genome and has transformed the process in which investigators search for regions of interest for a given cancer. In the past few years, the probe resolution has increased close to a thousand fold, between the first BAC arrays (2000-3000 probes) and the latest oligonucleotide arrays (1–2 million probes). This improved resolution allows for identification of micro-aberrations as well as more precise boundaries of known aberrations. Given that this technology yields a large number of potential candidates for further investigation, the challenge now is prioritization of this list to discover those that are functionally relevant. This involves integration of multiple data sources, preferably derived from the sample patients, via computational and statistical analysis. When preceded by careful experimental design and followed by thorough quantitative analysis, this approach is likely to bring tremendous new insights for understanding molecular basis of cancer.

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